

TN SKILL- PROJECT REPORT
On
FARM MANAGEMENT SYSTEM

Project work submitted to



THIRUVALLUVAR UNIVERSITY – VELLORE

In partial fulfillment of the requirement

For the award of the degree of

BACHELOR OF SCIENCE

IN

SOFTWARE COMPUTER SCIENCE

Submitted

by

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Farm Management System

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Farm Management System

User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose :

The main purpose of this project is to:

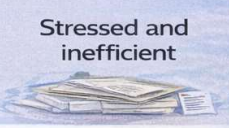
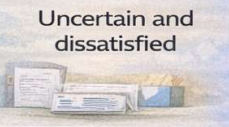
- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied

PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure
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Problem Statements – Farm Management System					
PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1		Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	
PS-2	 Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	 Stressed and inefficient
PS-3		Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	 Uncertain and dissatisfied
PS-4	 System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	 Concerned and insecure

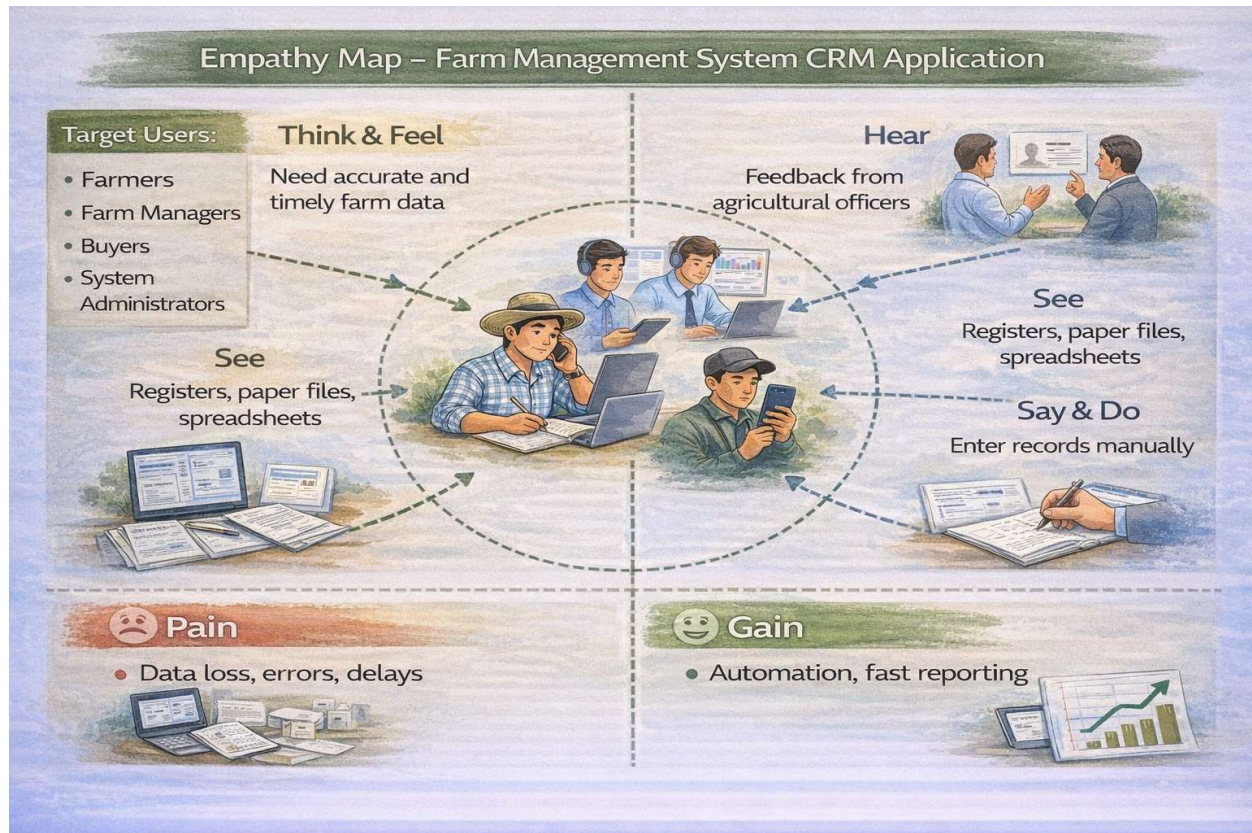
2.2 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- **Think & Feel:** Need accurate and timely farm data
- **Hear:** Feedback from agricultural officers
- **See:** Registers, paper files, spreadsheets
- **Say & Do:** Enter records manually
- **Pain:** Data loss, errors, delays
- **Gain:** Automation, fast reporting



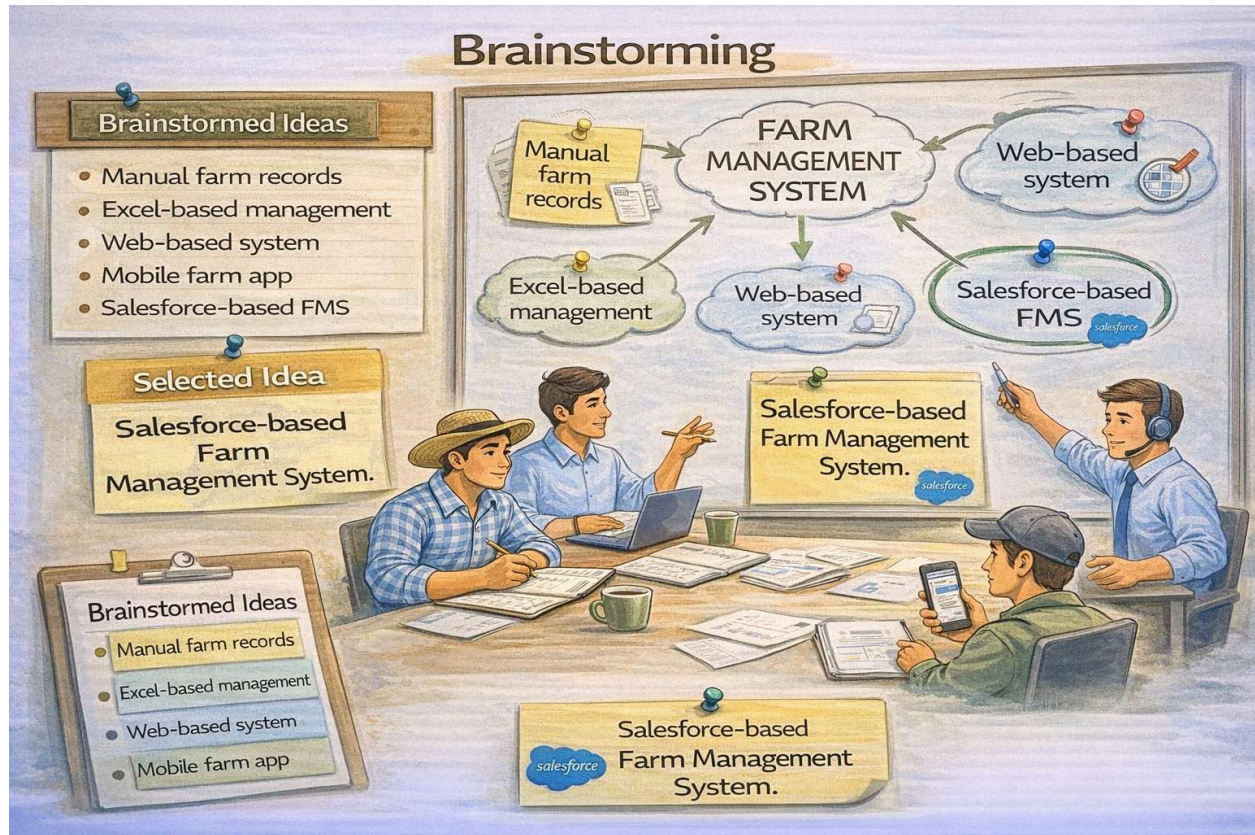
2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.



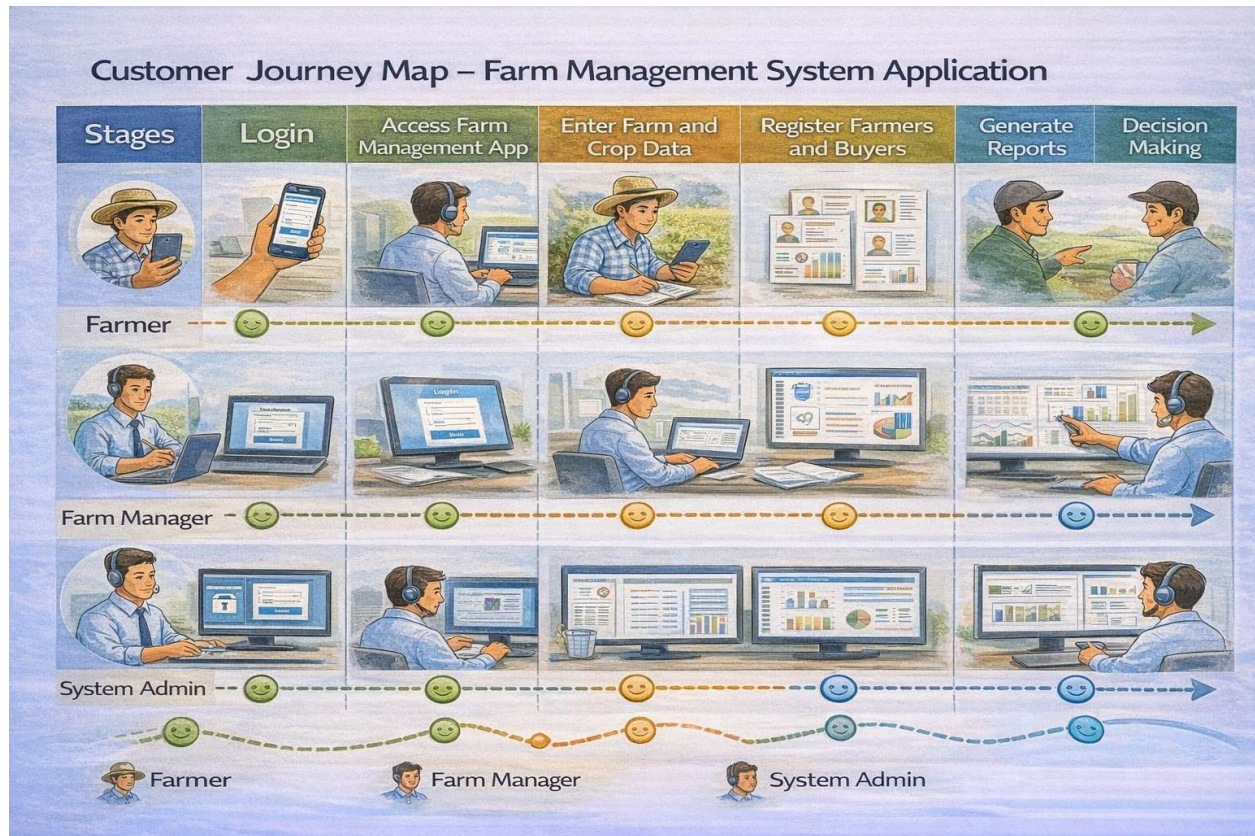
3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results

7. Decision Making



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
-------	-------------

FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation
FR-6	Dashboard Visualization
FR-7	User Role Management

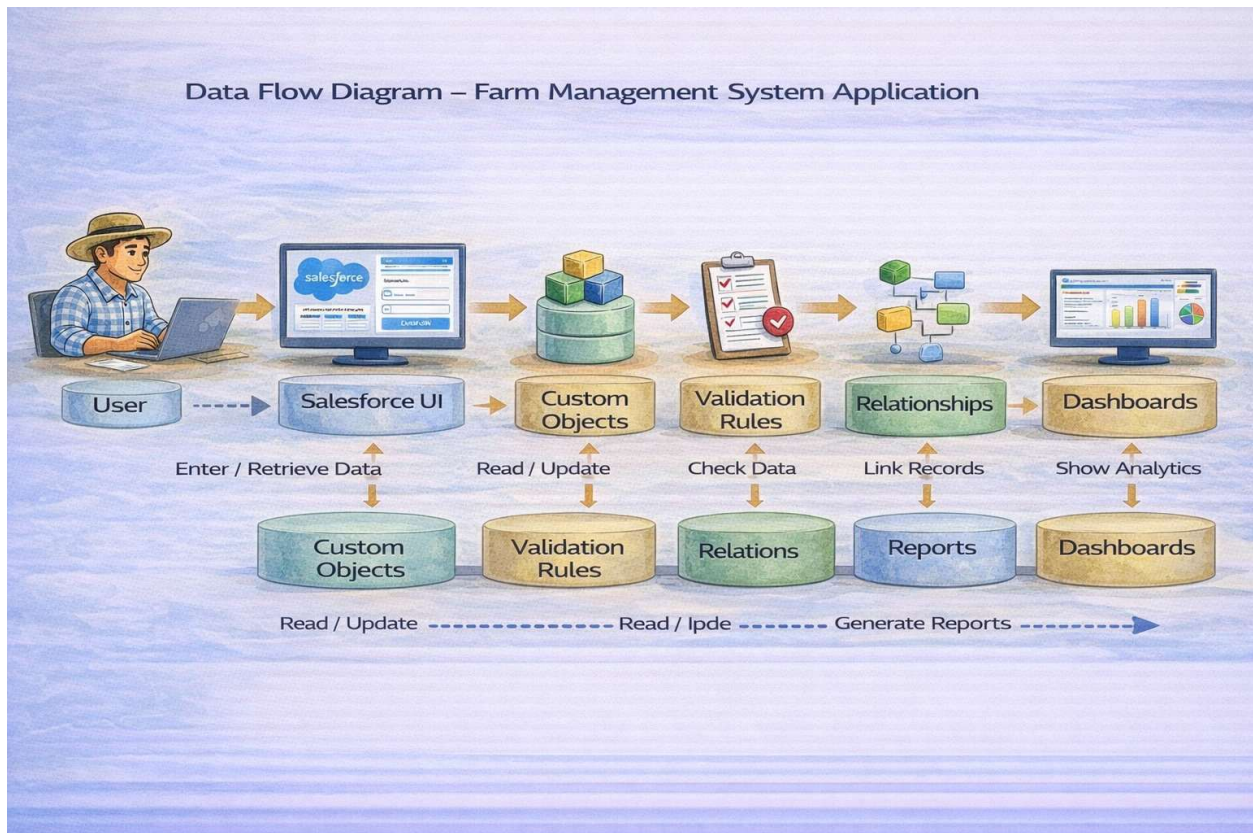
Non-Functional Requirements

NFR	Description	Example
Usability	Easy interface	Simple Lightning UI
Security	Data protection	Profiles & roles
Reliability	No data loss	Cloud backup
Performance	Fast access	Reports in <3 sec
Availability	24/7 access	Cloud hosting
Scalability	Handles growth	More farms/users

3.3 Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



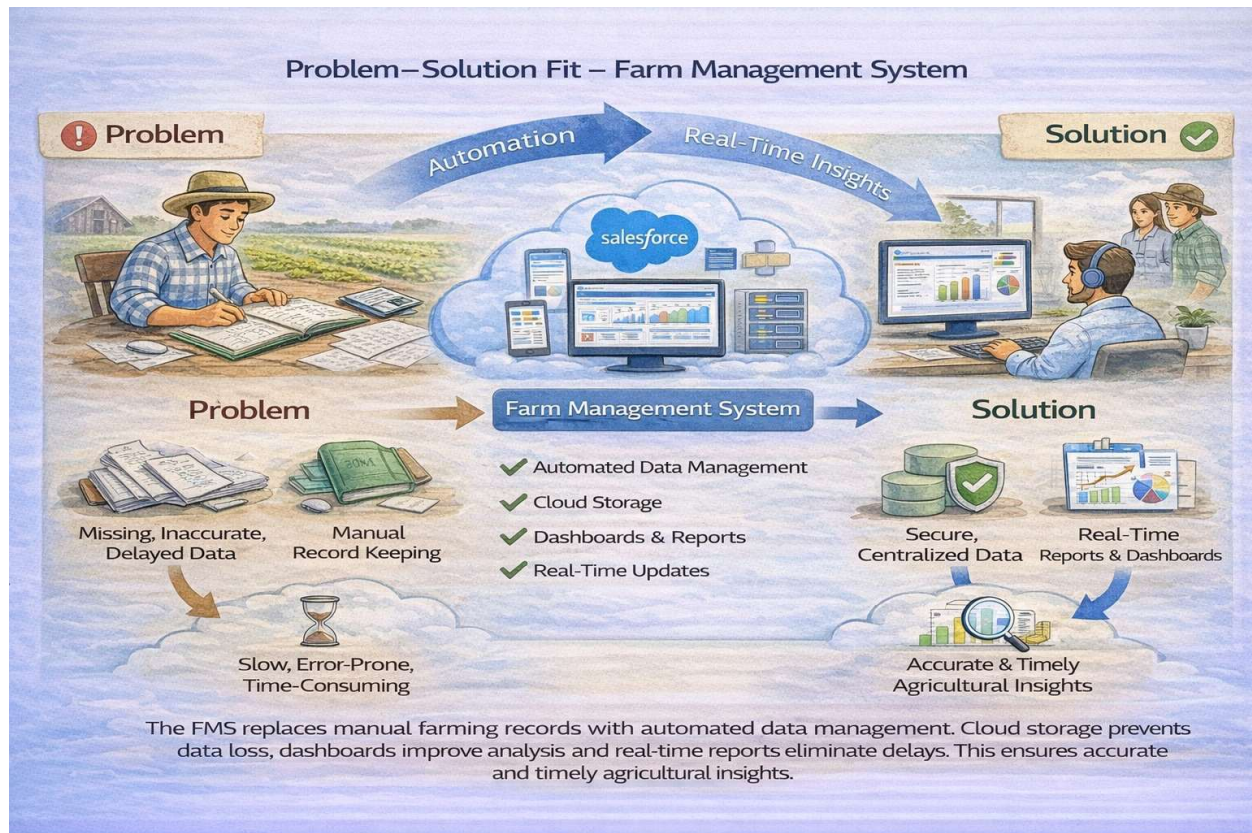
3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Validation Rules
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.



4.2 Proposed Solution

Functional Requirements

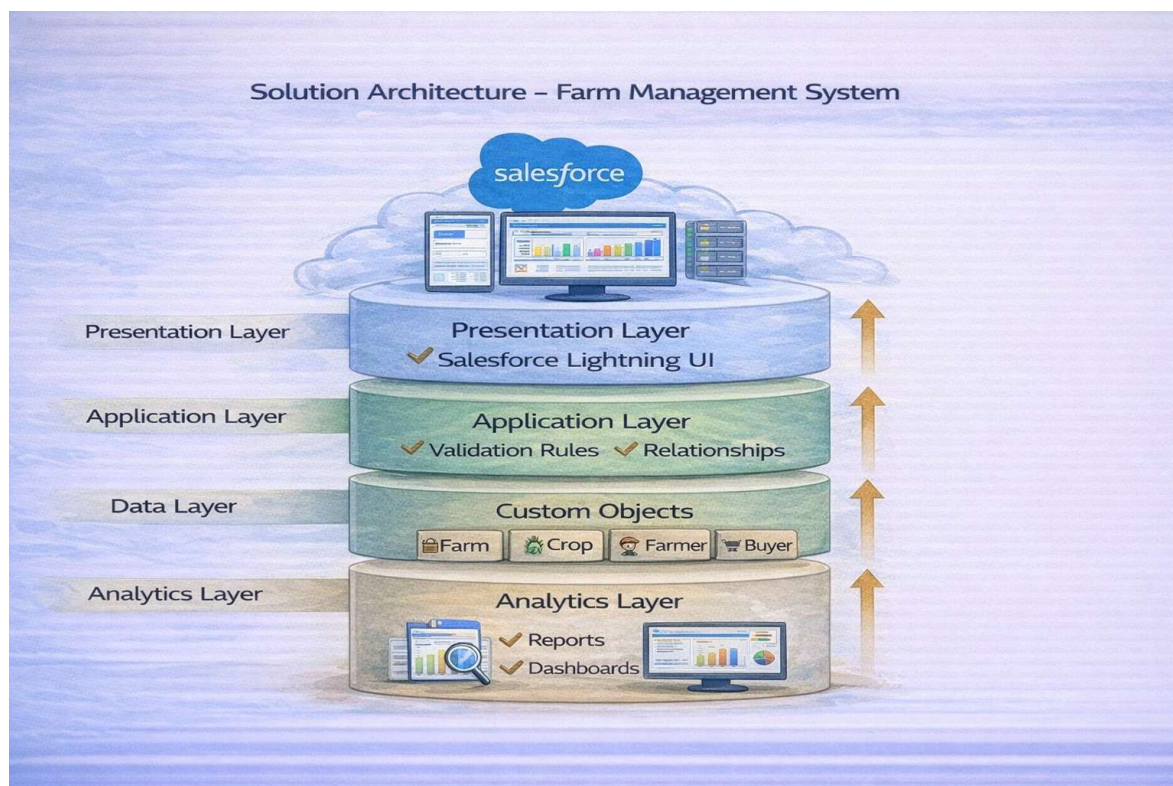
Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.
Solution	Salesforce-based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.

Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.es
Business Model	Digital farm management and agricultural data services platform.
Scalability	Supports multiple farms, users, and regions with cloud-based infrastructure.

4.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2a	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationships so that agricultural data is stored and managed efficiently.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records	3	High	Member4

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			are prevented.			
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase:

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

Salesforce has everything you need to run your business from anywhere. Using standard

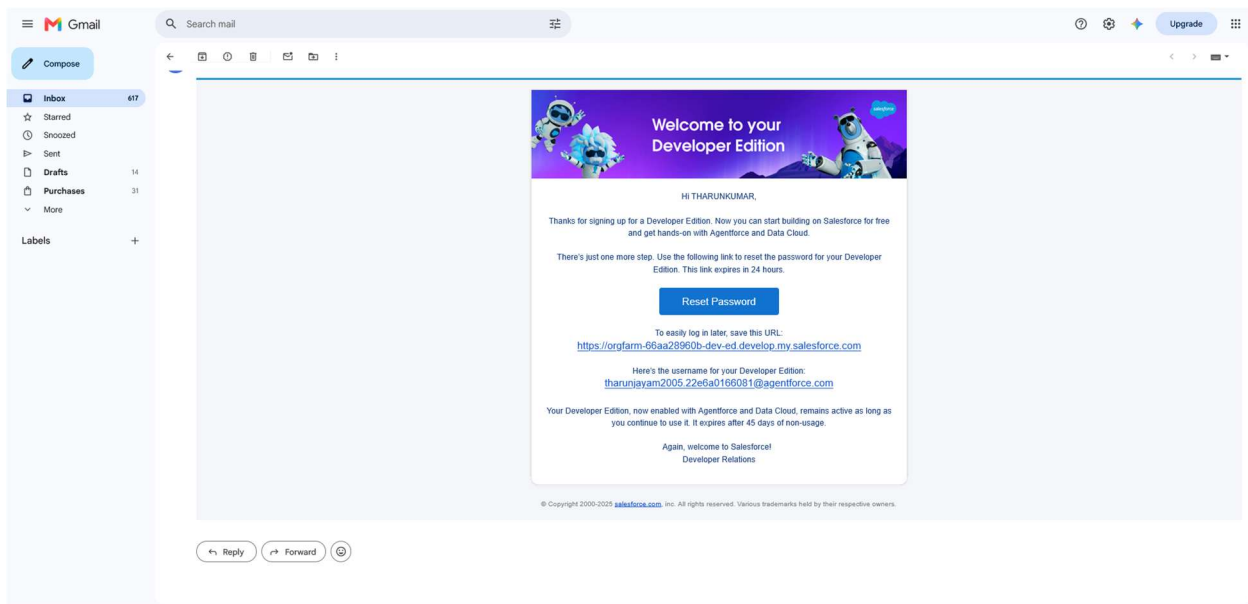
products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

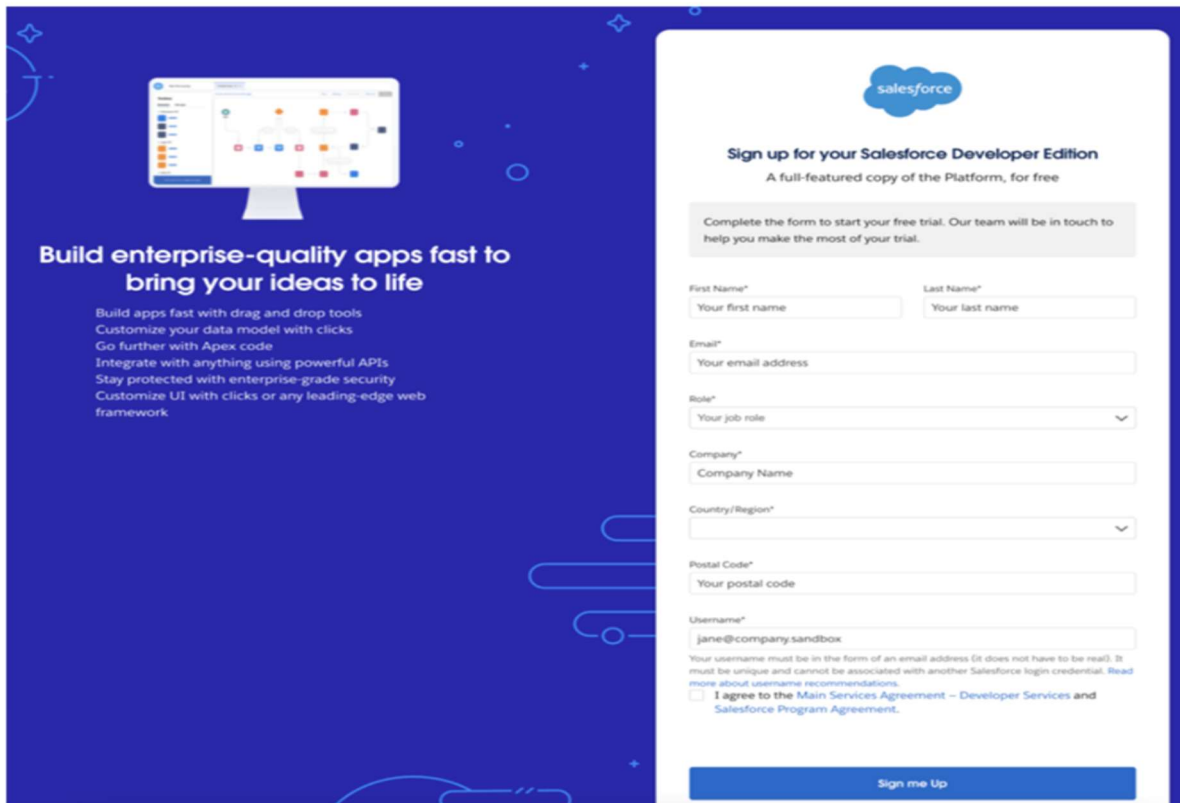
Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup)
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.
4. Click save.



5. Search login.salesforce.com



Build enterprise-quality apps fast to bring your ideas to life

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 Customize your data model with clicks
 Go further with Apex code
 Integrate with anything using powerful APIs
 Stay protected with enterprise-grade security
 Customize UI with clicks or any leading-edge web framework

Sign up for your Salesforce Developer Edition
 A full-featured copy of the Platform, for free

Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name* Your first name Last Name* Your last name

Email* Your email address

Role* Your job role

Company* Company Name

Country/Region*

Postal Code* Your postal code

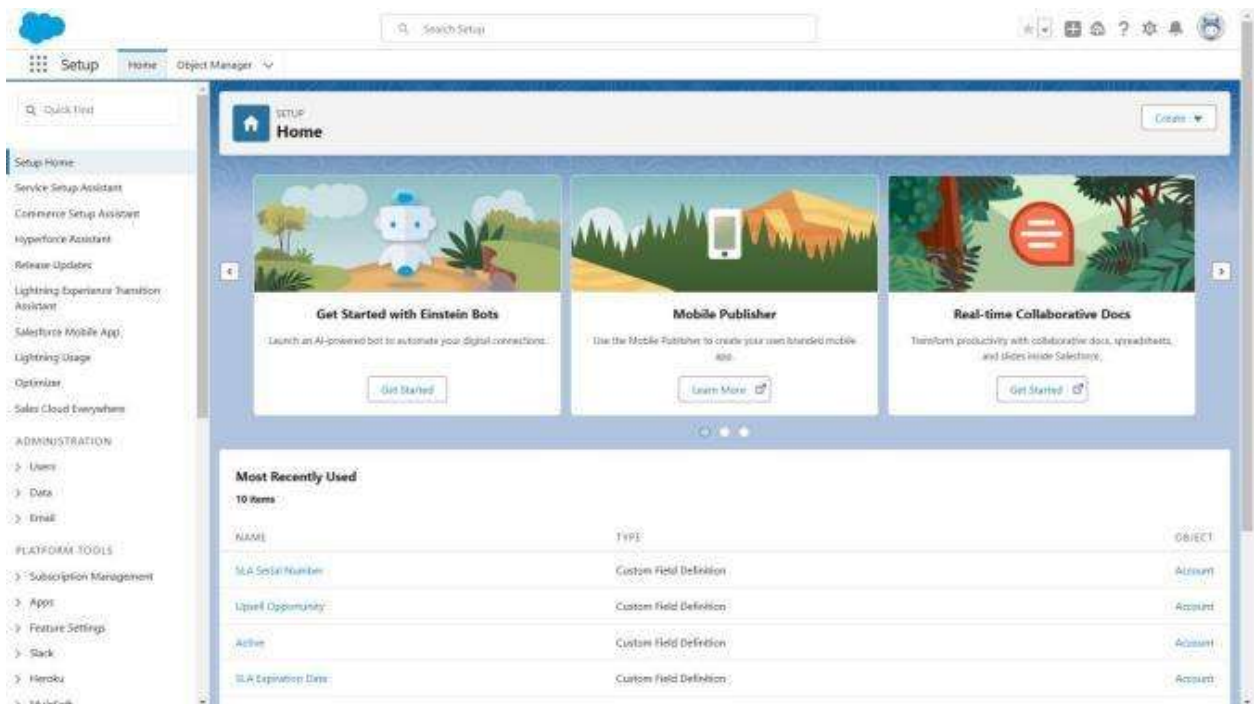
Username* jane@company.sandbox

Your username must be in the form of an email address (it does not have to be real). It must be unique and cannot be associated with another Salesforce login credential. [Read more about username recommendations.](#)

☐ I agree to the Main Services Agreement – Developer Services and Salesforce Program Agreement.

Sign me Up

6.By using username and password you can into the Salesforce org. The homepage will appear below



Setup Home

Get Started with Einstein Bots
 Launch an AI-powered bot to automate your digital connections.
[Get Started](#)

Mobile Publisher
 Use the Mobile Publisher to create your own branded mobile app.
[Learn More](#)

Real-time Collaborative Docs
 Transform productivity with collaborative docs, spreadsheets, and slides inside Salesforce.
[Get Started](#)

Most Recently Used
 10 Items

NAME	TYPE	OBJECT
SLA Serial Number	Custom Field Definition	Account
Liquor Opportunity	Custom Field Definition	Account
Active	Custom Field Definition	Account
SLA Expiration Date	Custom Field Definition	Account

Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

1.Farm

2.Crop

3.Farmer

4.Buyer

Activity 1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.

Developer Edition

Search...

Your Developer Edition Toolkit
Handpicked blogs, videos, and events to level up your skills.

Pro-Code Chunking of Data 360 Search Indexes for Agentforce
By Jitin Mehndiratta & Samartha Patil

Pro-Code Chunking of Data 360 Search Indexes for Agentforce
Learn how pro-code chunking with Data 360 Code Extension functions gives Agentforce the right context by preserving complex tables and conversations, masking PII, and improving search indexes.
[Read the guide](#)

Setup Menu

- Setup
- Data Cloud Setup
- Salesforce Go
- Agentforce Vibes
- Developer Console

29°C Sunny

Search

ENG IN

09:50 28-09-2026

Developer Edition

Search Setup

Setup Home Object Manager

Object Manager
53+ Items, Sorted by Label

Quick Find Schema Builder Create

Custom Object Custom Object from Spreadsheet

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED
Account	Account	Standard Object		
Activity	Activity	Standard Object		
Activity History	DevopsActivityLog	Standard Object		
Address	Address	Standard Object		
Agent Work	AgentWork	Standard Object		
Alternative Payment Method	AlternativePaymentMethod	Standard Object		
API Anomaly Event Store	ApiAnomalyEventStore	Standard Object		
Appointment Category	AppointmentCategory	Standard Object		
Appointment Invitation	AppointmentInvitation	Standard Object		

3. Fill in the label as " Farm ".

4. Fill in the plural label as " Farms".

5. Record name: Farm Name"

6. Select the data type as "Text".

The screenshot shows the Salesforce Setup interface in the 'Developer Edition'. The 'Object Manager' tab is selected, and the 'Farm' object is being edited. The 'Details' sidebar on the left lists various configuration options. The main area is titled 'Custom Object Definition Edit' and contains the following sections:

- Custom Object Information:** A warning note states, 'The singular and plural labels are used in tabs, page layouts, and reports. Be careful when changing the name or label as it may affect existing integrations and merge templates.' It includes fields for 'Label' (set to 'Farm'), 'Plural Label' (set to 'Farms'), and 'Object Name' (set to 'Farm').
- Description:** A large text area for the object's description.
- Context-Sensitive Help Setting:** Two radio buttons: 'Open the standard Salesforce.com Help & Training window' (selected) and 'Open a window using a Visualforce page'.

This screenshot shows the lower portion of the 'Custom Object Definition Edit' page for the 'Farm' object. The 'Data Type' is set to 'Text'. A warning message reads: 'Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.' The sections shown are:

- Optional Features:** Includes checkboxes for 'Allow Reports' (checked), 'Allow Activities' (unchecked), 'Track Field History' (unchecked), 'Allow in Chatter Groups' (unchecked), and 'Enable Licensing' (unchecked).
- Object Classification:** A note explains that when these settings are enabled, the object is classified as an 'Enterprise Application object'. It includes checkboxes for 'Allow Training' (checked), 'Allow Bulk API Access' (checked), and 'Allow Streaming API Access' (checked).
- Deployment Status:** Includes radio buttons for 'In Development' (unchecked) and 'Deployed' (checked).
- Search Status:** Includes a checkbox for 'Allow Search' (checked).

7. In the Optional Features section, select Allow Reports and Track Field History.

8. In the Deployment Status section, ensure Deployed is selected.

9. In the Search Status section, select Allow Search.

10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.

11. Leave everything else as is, and click Save

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop
 - c. Plural Label: Crops
 - d. Record Name: Crop Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox g. Check the Allow Search checkbox
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout i. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Farmer
 - c. Plural Label: Farmers
 - d. Record Name: Farmer Name
 - e. Select the data type as "Text".

- f. Check the Allow Reports checkbox.
- g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
- i. Click Save.

Activity 4:

Creation of Buyer object:

- 1. Click on the gear icon and then select Setup.
- 2. Click on the object manager tab just beside the home tab.
- 3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Buyer
 - c. Plural Label: Buyers
 - d. Record Name: Buyer Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
 - i. Click Save.

Milestone 3 -

Tab: Tab

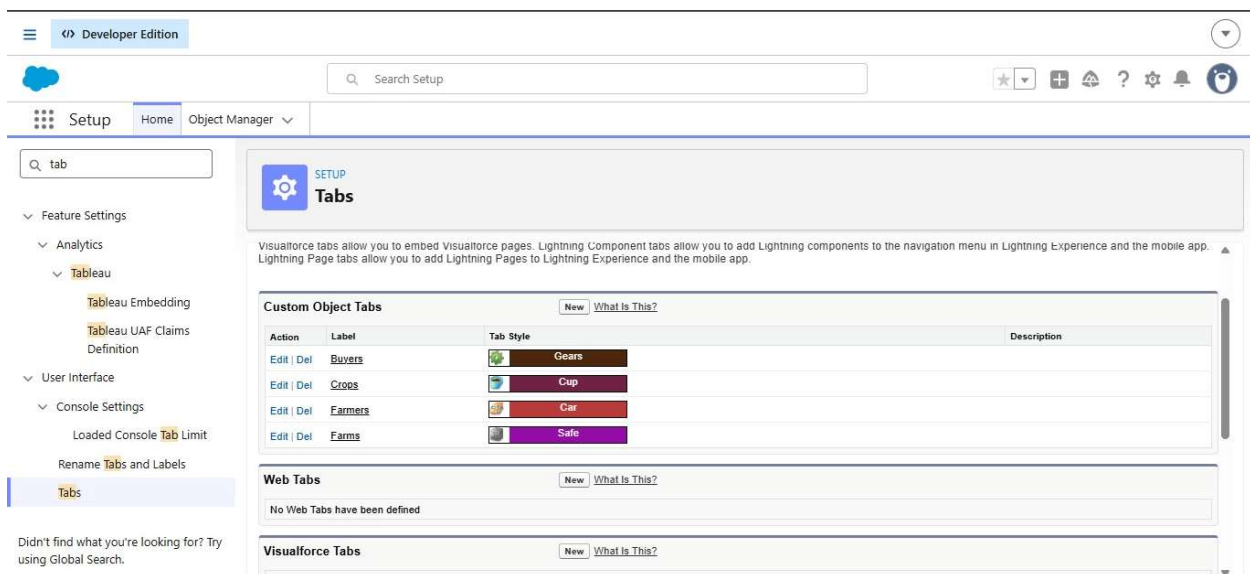
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



The screenshot shows the Salesforce Setup interface. The left sidebar contains the navigation menu with 'Setup' selected. The main content area displays the 'Tabs' configuration page. The 'Custom Object Tabs' section shows a table of existing tabs:

Action	Label	Tab Style	Description
Edit Del	Buyers		Gears
Edit Del	Crocs		Cup
Edit Del	Farmers		Car
Edit Del	Farms		Safe

Below the table, there are sections for 'Web Tabs' and 'Visualforce Tabs', both of which are currently empty.

2. Under custom object tabs, click New.



3. For Object, select Farm.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

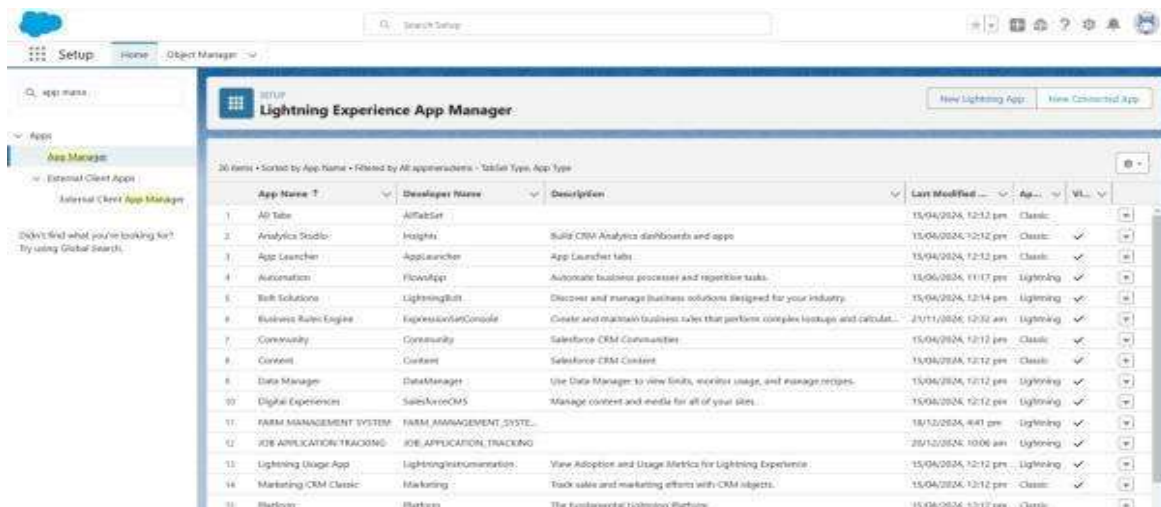
Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app



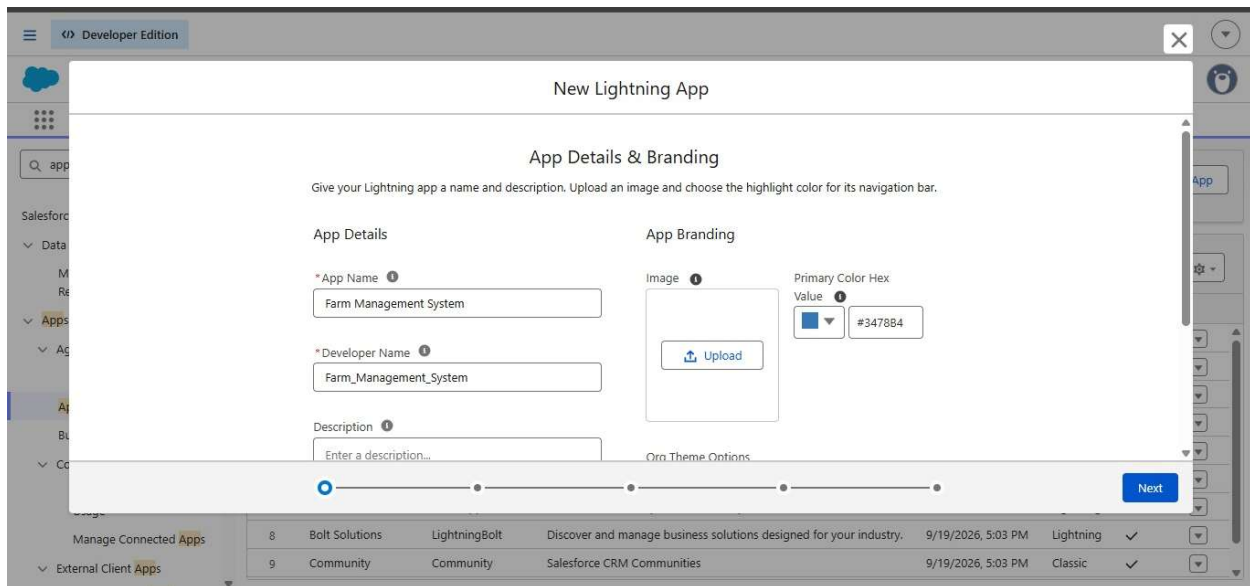
- From Setup, enter App Manager in the Quick Find and select App Manager.
- Click New Lightning App. Farm Management System as the App Name, then click Next
- Under App Options, leave the default selections and click Next.
- Under Utility Items, leave as is and click Next.
- From Available Items, select Farm, Crop, Farmer, Buyer

Reports, and Dashboards and move them to Selected Items. Click Next. ● From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.

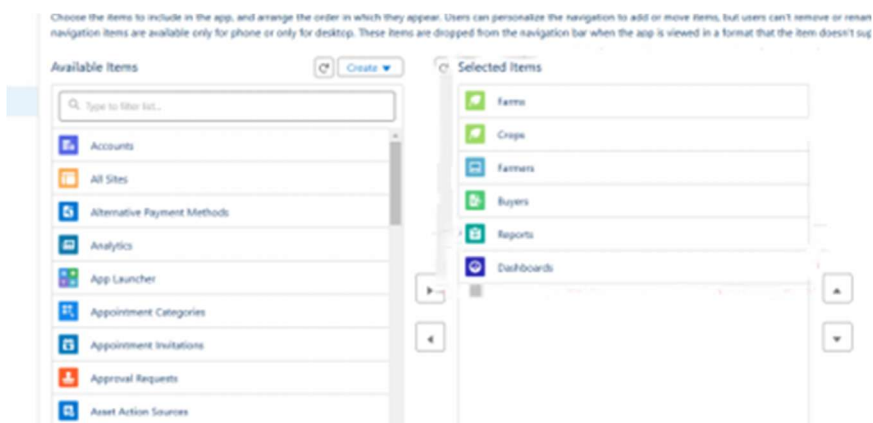
- To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

Note:

1. App Launcher-Displays available apps.
2. App Name-Displays the current selected app.



3. Navigation menu-Displays the tabs available inside the app.



Milestone 5 -

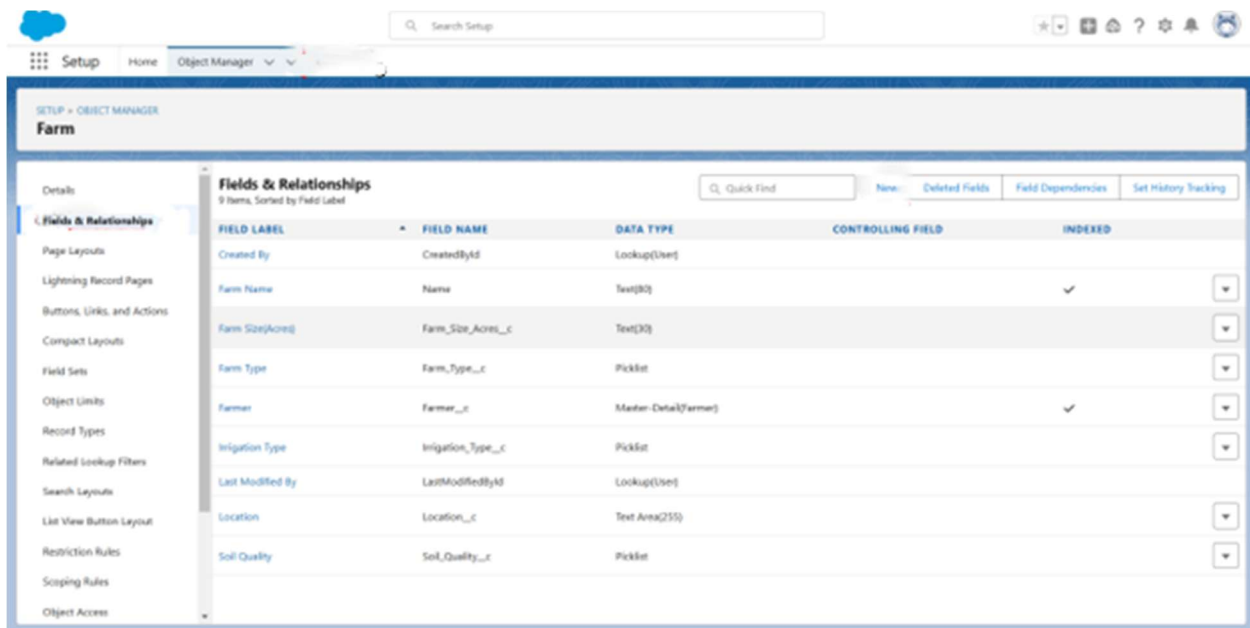
Fields and Relationship: What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:

Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New



5. Choose Data type as a picklist, Click next

6. Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)

7. Next, Next Click Save.

The screenshot shows the Salesforce Setup interface in the Developer Edition. The top navigation bar includes the Setup menu, Home, and Object Manager. The main content area is titled 'Farm' and shows the 'Custom Object Definition Edit' page. The left sidebar lists various setup options under 'Details', including Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, and Related Lookup Filters. The main panel displays the 'Custom Object Information' section, which includes fields for Label (Farm), Plural Label (Farms), Object Name (Farm), and a Description field. There are also checkboxes for 'Starts with vowel sound' and 'Context-Sensitive Help Setting'. The page includes 'Save', 'Save & New', and 'Cancel' buttons at the top right.

Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

1. Next we Create another Picklist Field, Enter field label (Soil Quality) enter values (Organic, Cover Crops, Composting)
2. Again we Create Picklist field, Field Label (Irrigation type), Enter Values (Sprinkler irrigation, Drip irrigation, localized irrigation)
3. Choose data type as a text, enter field label (Location) 4. Choose data type as text, enter field label (size)
5. Create a Master Detail relationship with Farmer Object.

Activity 2:

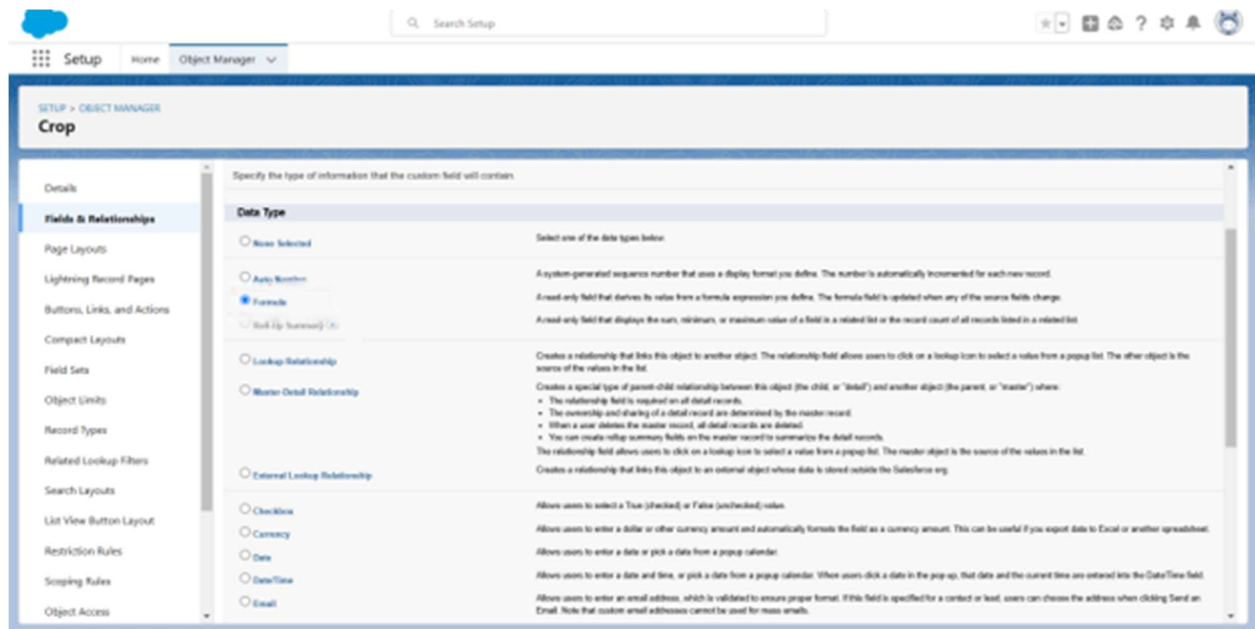
Creation of fields for the Crop objects:

1. Choose Data type As a text, Enter Field Label name (Crop Name), Next save.
2. Select Data type as date, Enter Field label (Planting Date), then save.
3. Again we select data type as a date, Enter field label (Harvest date), save
4. Next We Create Lookup relationship on crop object related to Farm Object.

5. We select a data type as currency, Enter field Label (Actual Yield), Next Save it.

6. Select Data type as a Number, enter Field Label (Total Acres Harvested), Enter Maximum Length(20) and save.

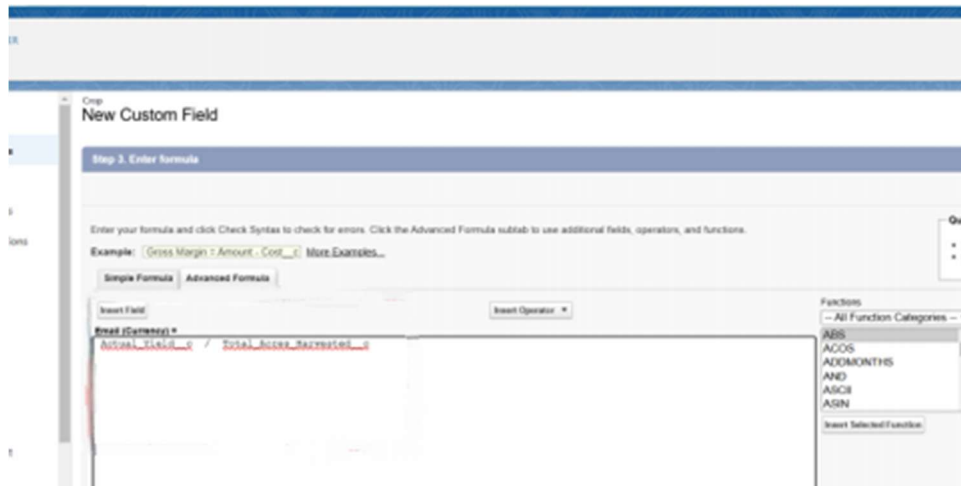
7. Now We create Formula Field For That Select data Type As a Formula, Click Next.



The screenshot shows the Salesforce Setup interface for the 'Crop' object. The left sidebar contains a navigation menu with options like 'Details', 'Fields & Relationships', 'Page Layouts', 'Lightning Record Pages', 'Buttons, Links, and Actions', 'Compact Layouts', 'Field Sets', 'Object Limits', 'Record Types', 'Related Lookup Filters', 'Search Layouts', 'List View Button Layout', 'Restriction Rules', 'Scoping Rules', and 'Object Access'. The main content area is titled 'Crop' and has a sub-header 'Specify the type of information that the custom field will contain'. Under the 'Data Type' section, several options are listed with radio buttons. The 'Formula' option is selected and highlighted. Other options include 'None Selected', 'Auto Number', 'Lookup Relationship', 'Master-Detail Relationship', 'External Lookup Relationship', 'Checkbox', 'Currency', 'Date', 'DateTime', and 'Email'. Each option has a brief description of its functionality.

8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.



Activity 3:

Creation of fields for the Farmer objects:

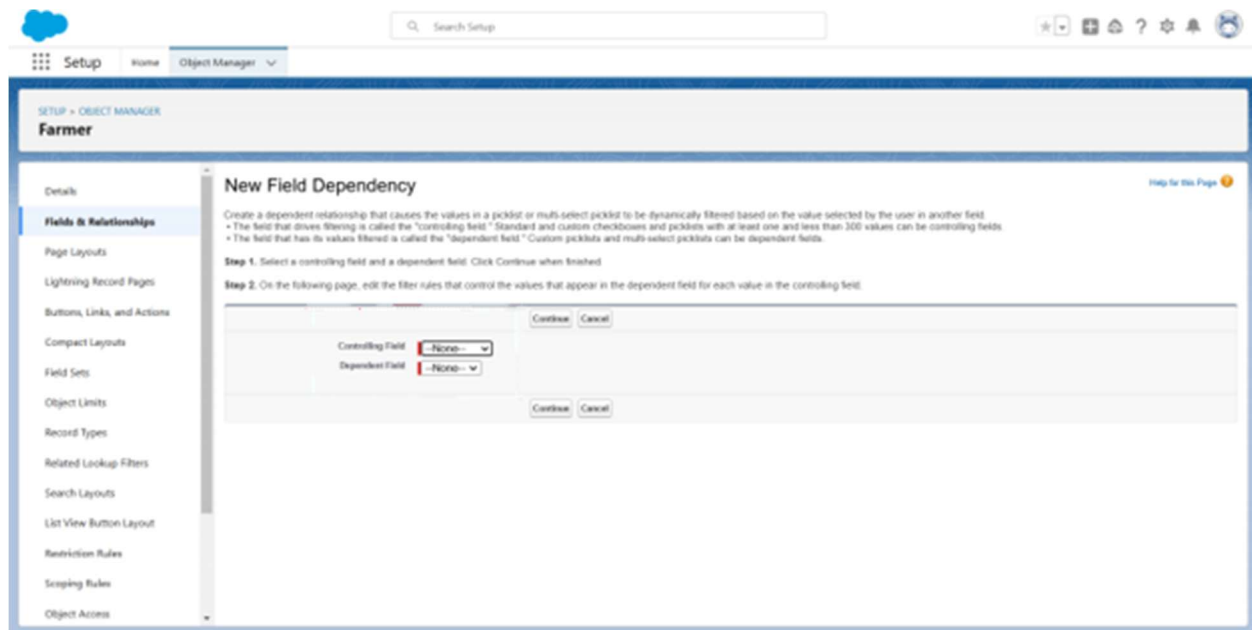
1. Select Data type Phone, Enter field label (Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label (State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Creation of field dependency :

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer object, select Fields & Relationships, On Right side We have Field Dependency option, select that option.
2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.

5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.

6. Click Preview, then test the dependency by selecting a different State and viewing the associated Values available for Particular state.



7. Click Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1. Select Data type as a Phone, Enter a Field Label(Contact), Save.
2. Select Data type as an Email, Enter Field Label(Email), Next save.
3. Select Data type Text area, Enter Field Label(Address), Save it.
4. Select Data type Picklist, Field Label(Buyer Type), Enter values (Wholesale, Retail), Click Save.

Milestone 6- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in

salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab

settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.

Developer Edition

Search Setup

Setup Home Object Manager

prof

Users

Profiles

Didn't find what you're looking for? Try using Global Search.

SETUP Profiles

Profiles

All Profiles Edit Delete Create New View

New Profile

Action	Profile Name	User License	Custom
Edit Clone	Analytics Cloud Integration User	Analytics Cloud Integration User	☐
Edit Clone	Analytics Cloud Security User	Analytics Cloud Integration User	☐
Edit Clone	Anypoint Integration	Identity	☐
Edit Clone	Authenticated Website	Authenticated Website	☐
Edit Clone	Authenticated Website	Authenticated Website	☐
Edit Del ...	B2B Reordering Portal Buyer Profile	External Apps Login	☒
Edit Clone	Chatter External User	Chatter External	☐
Edit Clone	Chatter Free User	Chatter Free	☐

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Developer Edition

Search Setup

Setup Home Object Manager

prof

Users

Profiles

Didn't find what you're looking for? Try using Global Search.

SETUP Profiles

Profile Custom: Sales Profile

Users with this profile have the permissions and page layouts listed below. Administrators can change a user's profile by editing that user's personal information.

If your organization uses Record Types, use the Edit links in the Record Type Settings section below to make one or more record types available to users with this profile.

[Login IP Ranges](#) |
 [Enabled Apex Class Access](#) |
 [Enabled Visualforce Page Access](#) |
 [Enabled External Data Source Access](#) |
 [Enabled Named Credential Access](#) |
 [Enabled External Credential Principal Access](#) |
 [Enabled Custom Metadata Type Access](#) |
 [Enabled Custom Settings Definitions Access](#) |
 [Enabled Flow Access](#) |
 [Enabled Service Presence Status Access](#) |
 [Enabled Custom Permissions](#)

Profile Detail

Edit Clone Delete View Users

Name	Custom: Sales Profile
User License	Salesforce
Description	
Created By	OroFarm EPIC, 9/18/2026, 5:40 AM
Modified By	Devakumar S, 9/25/2026, 12:30 AM

Page Layouts

Standard Object Layouts	Location Group Assignment
Global Global Layout	Location Group Assignment Layout

5. Click save.

Activity 2:

Create a custom profile-2:

Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box

7. Select profiles (Standard user).

8. Click clone.

Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option

The screenshot displays the Salesforce Setup interface. The top navigation bar includes the 'Developer Edition' label, a search bar, and various utility icons. The left sidebar shows the 'Setup' menu with 'Roles' highlighted under the 'Users' section. The main content area is titled 'SETUP Roles' and features a 'Sample Role Hierarchy' section. This section includes a dropdown menu for 'Territory-based Sample' and a hierarchical diagram. The diagram shows a top-level role 'Executive Staff' (CEO, President, CFO, VP, Sales) which branches into three director roles: 'Western Sales Director', 'Eastern Sales Director', and 'International Sales Director'. Each director role further branches into specific sales representative roles: 'CA Sales Rep', 'OR Sales Rep' for Western; 'NY Sales Rep', 'MA Sales Rep' for Eastern; and 'Asian Sales Rep', 'European Sales Rep' for International. To the right of the diagram, there are three text boxes providing details for each level: the top box for Executive Staff, the middle box for Directors, and the bottom box for Sales Representatives. At the bottom of the page, there is a 'Set Up Roles' button and a checkbox labeled 'Don't show this page again'.

3. Below the CEO click on add role and enter the label name as a " Farm Manager " and role name will be Automatically populated and click on save.

Milestone 8-User :

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in

Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find

salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.

The screenshot shows the Salesforce Setup interface. The left sidebar contains a search bar with 'user' entered and a list of navigation items including 'Users'. The main content area is titled 'SETUP Users' and includes instructions on creating, viewing, and managing users. Below this, there is a table of existing users with columns for Action, Full Name, Alias, Username, Role, Active status, and Profile. The table lists five users: Chatter Expert, EPIG_QrgFarm, S_Devakumar, User_Integration, and User_Security. At the bottom of the table, there are buttons for 'New User', 'Reset Password(s)', and 'Add Multiple Users'.

Action	Full Name	Alias	Username	Role	Active	Profile
☐ Edit	Chatter Expert	Chatter	chattv.00d9k00000ahzfluag.0naksts1h4s@chatter.salesforce.com		✓	Chatter Free User
☐ Edit	EPIG_QrgFarm	OEPIC	epic.9c4e7a84cc73@orgfarm.salesforce.com		✓	System Administrator
☐ Edit	S_Devakumar	jsd	jsdevakumar2007.7262fc1a968cc@agentforce.com		✓	System Administrator
☐ Edit	User_Integration	integ	integration@00d9k00000ahzfluag.com		✓	Analytics Cloud Integration User
☐ Edit	User_Security	sec	insightssecurity@00d9k00000ahzfluag.com		✓	Analytics Cloud Security User

3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as “Sales Department”, and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

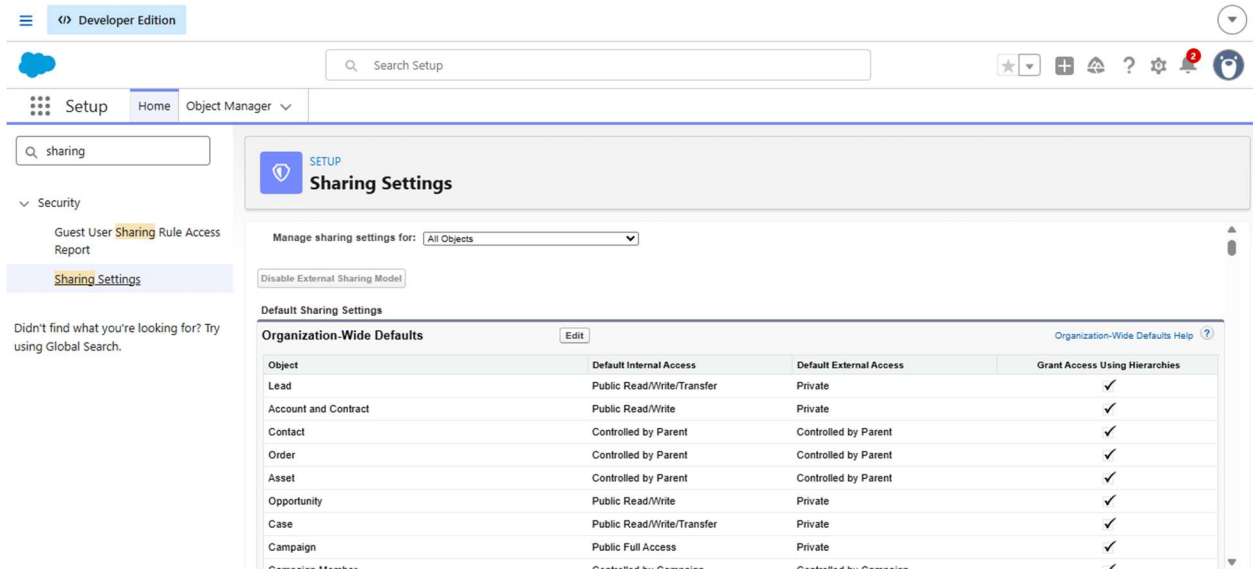
Types of sharing rules,

•Owner-based Sharing Rules •Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

- 1) Goto Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with the section select roles and in that select Farm Manager.
- 8) And in the section of select the level of access for the users give the access Read/Write.
- 9) And save the rule.



Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1: Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.

The screenshot shows a web application interface for farm management. A modal window titled "Edit Kesava" is open, displaying a form with the following fields: "Farmer Name" (text input with "Kesava"), "Contact Number" (text input with "7982627728"), "State" (dropdown menu with "Andhra Pradesh"), "District" (dropdown menu with "Chittoor"), "Village" (dropdown menu with "Vendaballe"), and "Farm Size(Acres)" (text input with "15"). The "Owner" field shows "Sushanth Jagann". At the bottom of the modal are three buttons: "Cancel", "Save & Notify", and "Save". The background interface includes a sidebar with "Recently Viewed" farms and a top navigation bar with "Import", "Change Owner", and "Assign Label" buttons.

Milestone 11-Reports:

What are Reports?

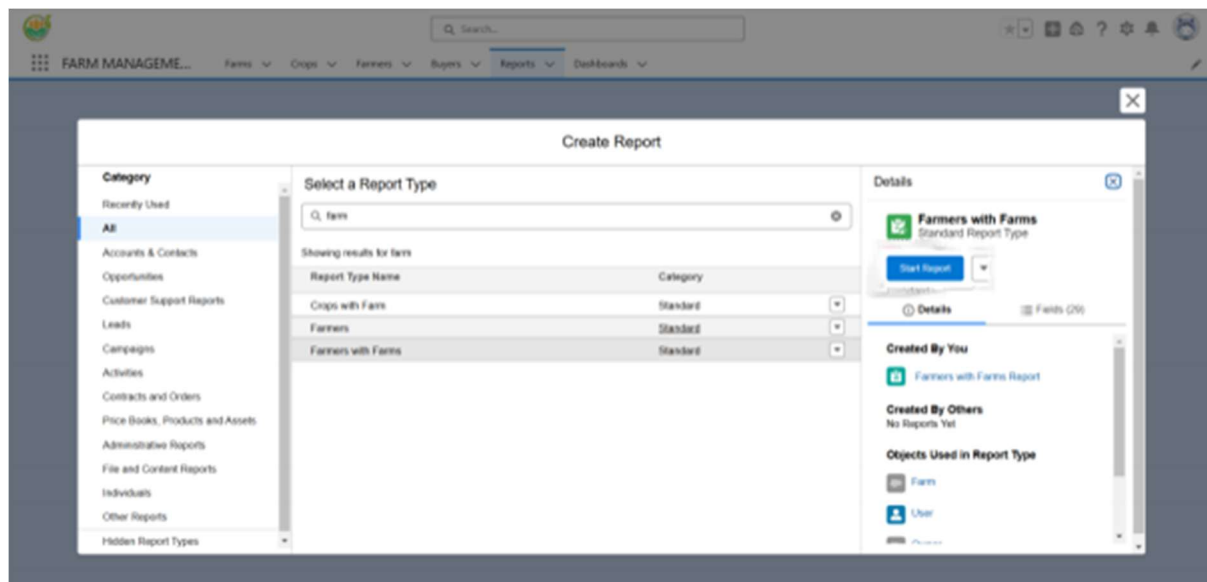
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

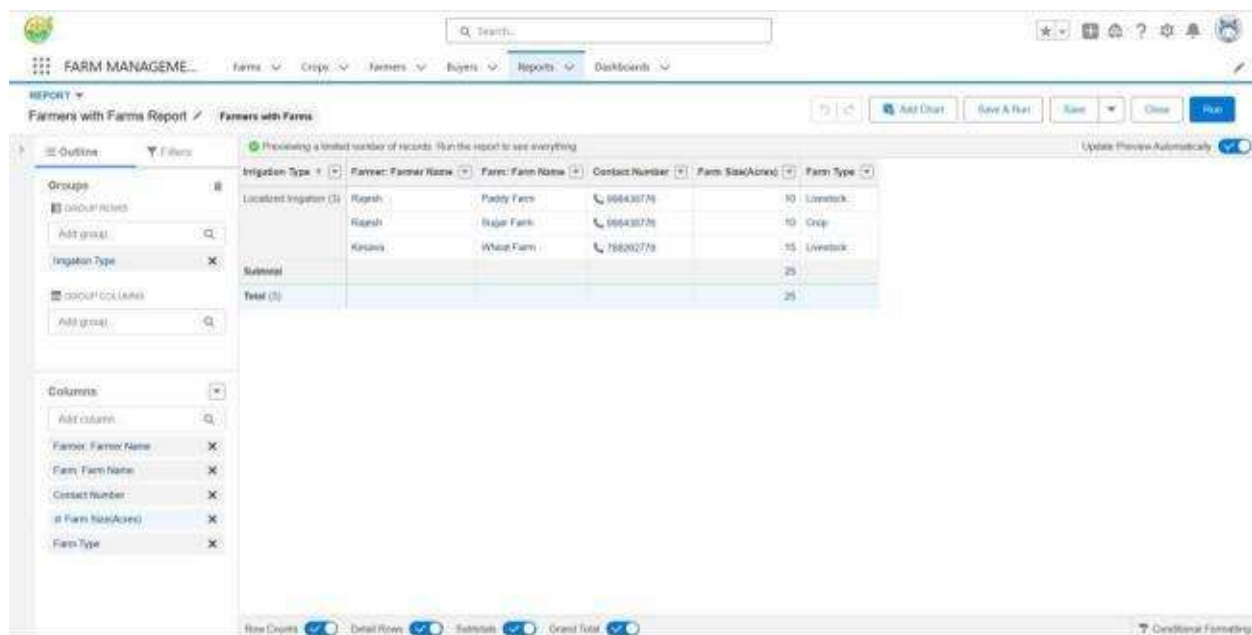
Create a report:

Create a report that displays rating of the account and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.



- 3) In the details section select the option start report.
- 4) In Outline Click on Column Select Which Field you want to display in the Report.
- 5) Click Group,Select Which Fields You Want to Group.
- 6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.



Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

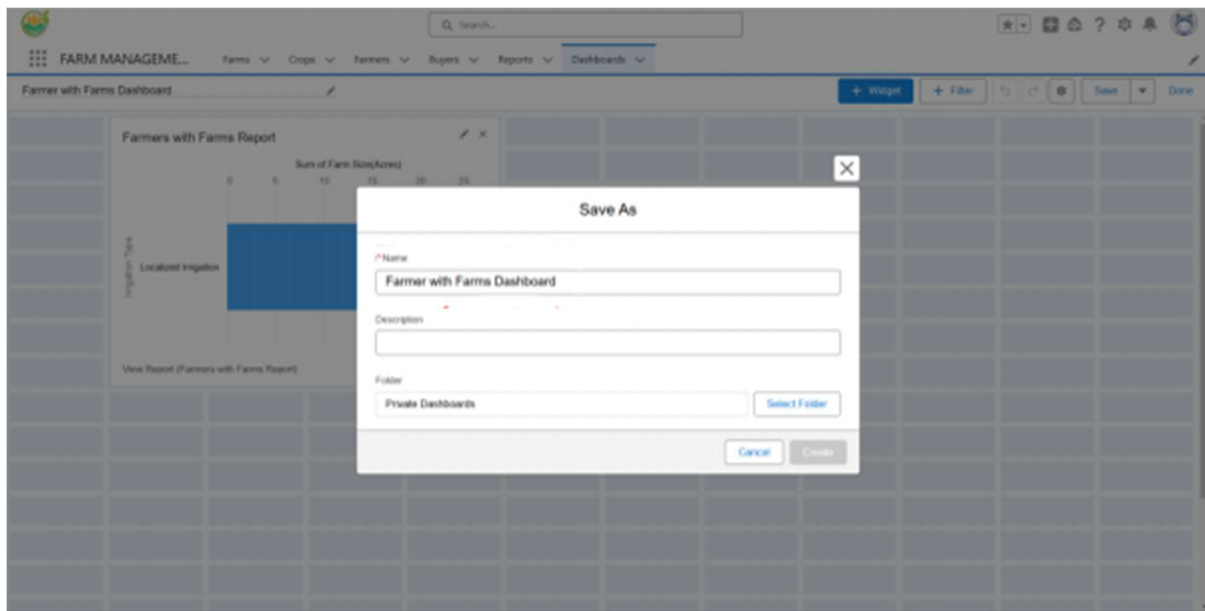
Milestone 12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

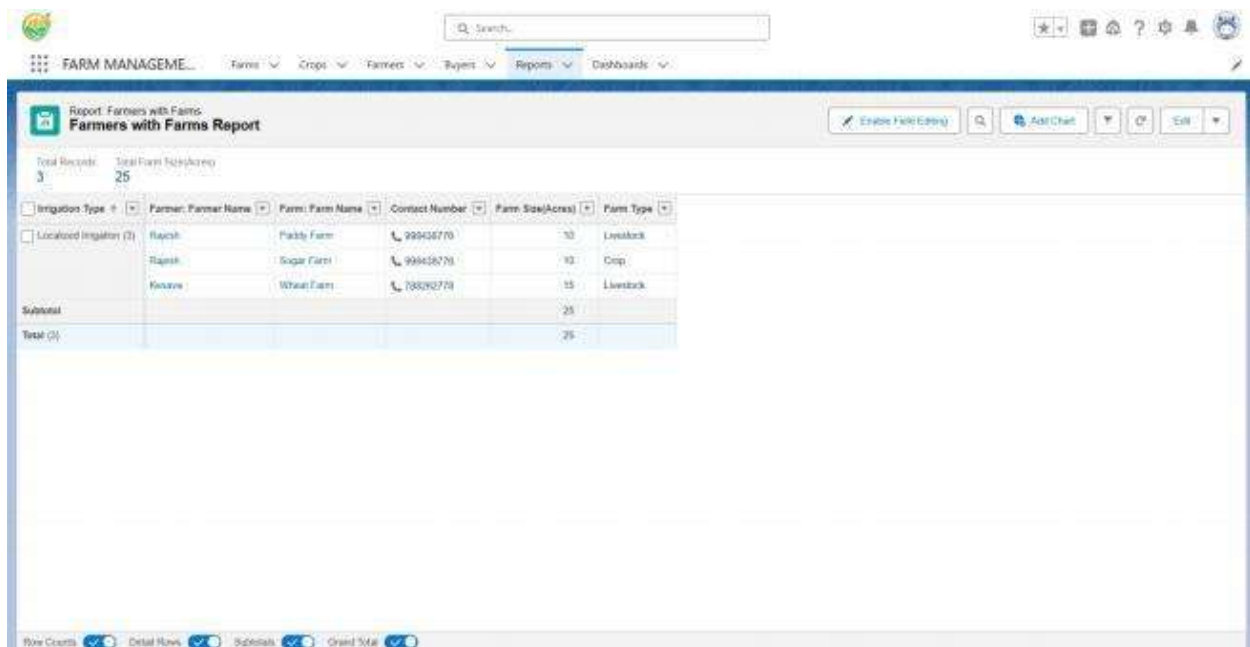
1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done



Milestone-13: View Report and Dashboard:

Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records

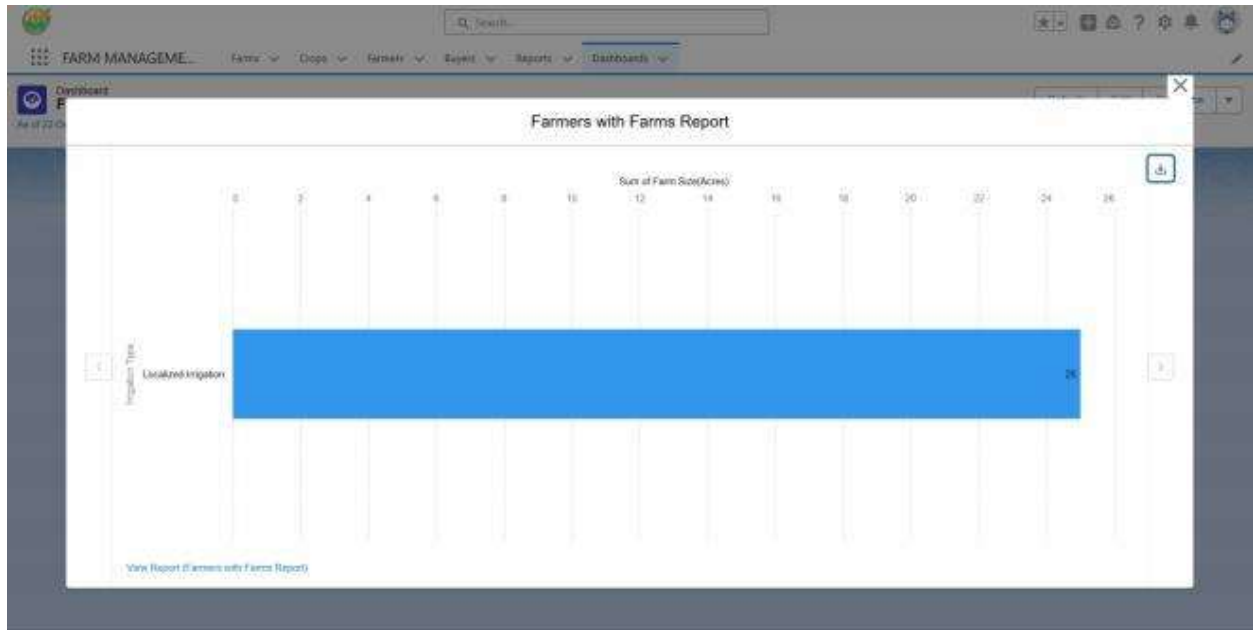


The screenshot displays the 'Farmers with Farms Report' interface. At the top, there is a search bar and navigation tabs for Farms, Crops, Farmers, Reports, and Dashboard. The 'Reports' tab is selected. Below the navigation, the report title 'Farmers with Farms' is shown. A table lists farm records with columns for Irrigation Type, Farmer Name, Farm Name, Contact Number, Farm Size (Acres), and Farm Type. The table includes three data rows and a total row. At the bottom, there are checkboxes for 'Row Counts', 'Detail Rows', 'Schedules', and 'Grand Total'.

Irrigation Type	Farmer: Farmer Name	Farm: Farm Name	Contact Number	Farm Size(Acres)	Farm Type
Localized Irrigation (3)	Rajesh	Paddy Farm	999436770	10	Livestock
	Rajesh	Sugar Farm	999436770	10	Crop
	Kavase	Wheat Farm	788262770	15	Livestock
Subtotal				25	
Total (3)				25	

Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

Thank You